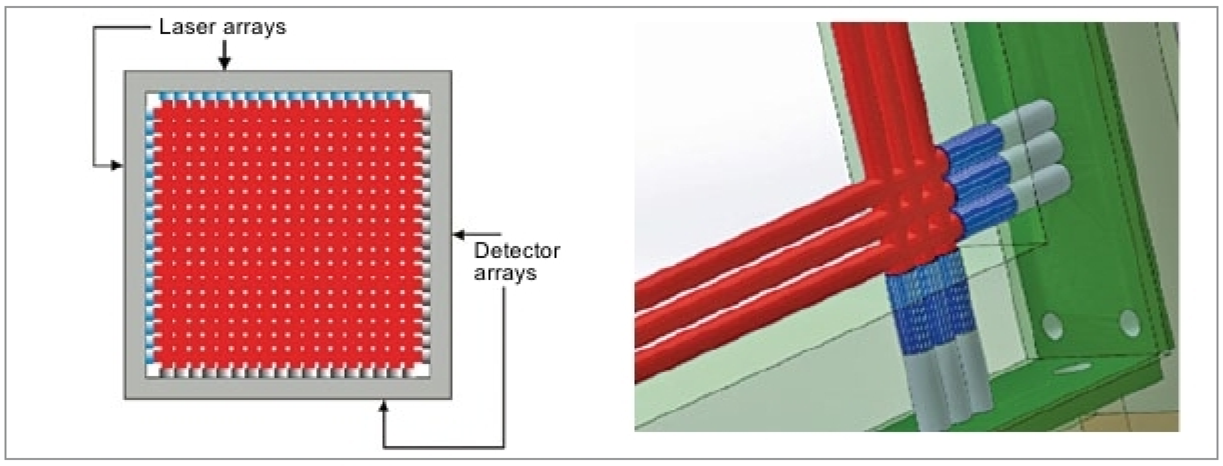


Fig. 14: Creating two-dimensional laser screen by using separate transmitter and receiver pairs

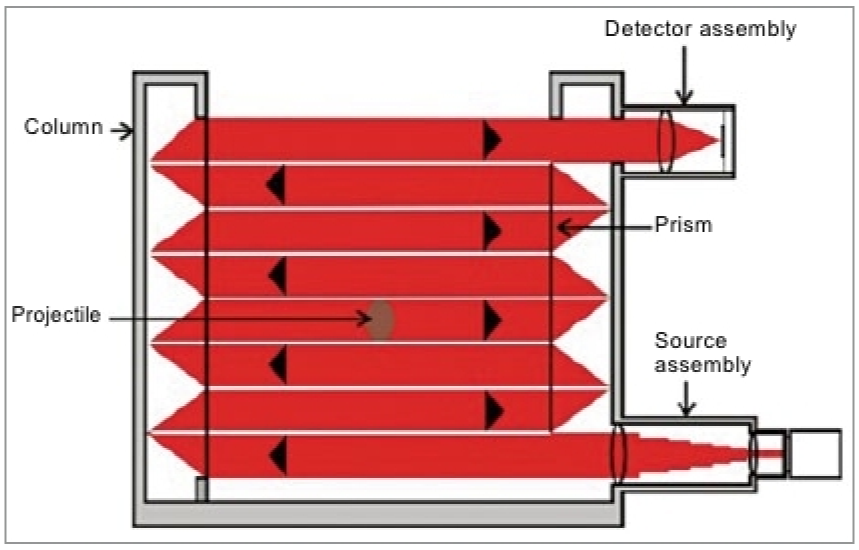


Fig. 15: Creating two-dimensional laser screen by employing a transmitter, a receiver and multiple retro-reflectors for each plane

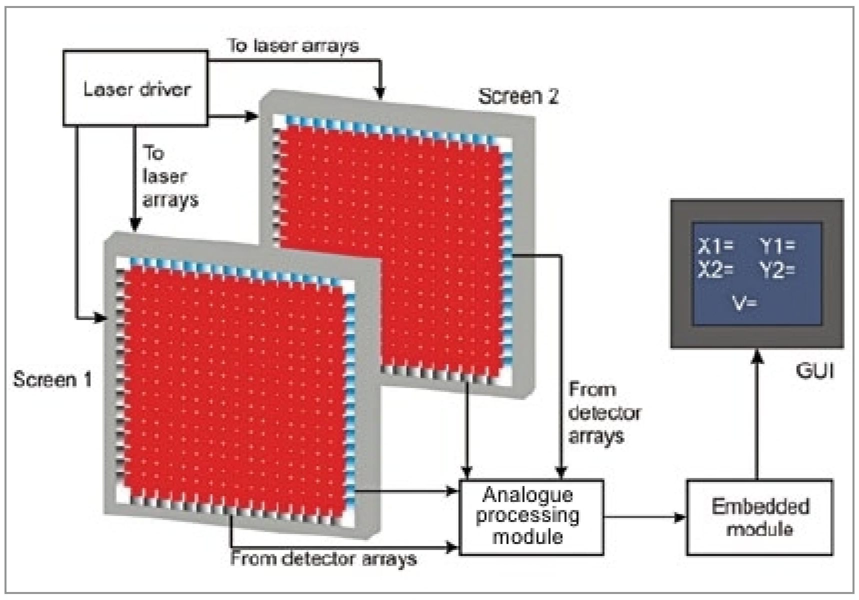


Fig. 16: Block schematic arrangement of different sub-systems comprising laser velocity screen instrumentation

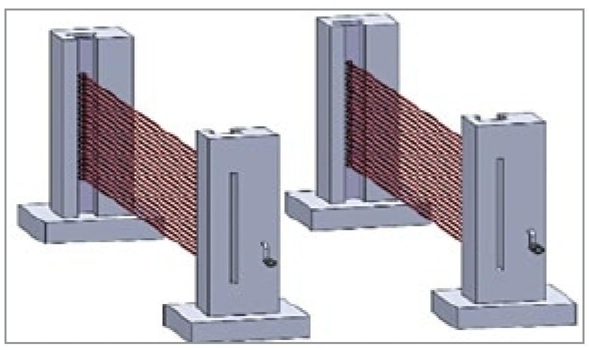

Fig. 17: Representative setup of laser screens

passes through the first laser screen, it cuts across one or more laser beams parallel to x-axis and one or more laser beams parallel to y-axis, giving x and y coordinates of projectile at that point. The resolution with which x and y coordinates of the projectile can be determined depends on the number of elements in the linear array.

Resolution for a given array size can be further improved by using a suitable interpolation algorithm. The second screen, spatially apart from the first screen, can similarly give x-y coordinates of the projectile at that location. Thus, we have x-y coordinates of the travelling projectile at two known positions along the path of the projectile. This information can be processed to determine the direction of movement, aiming accuracy and velocity.

Velocity is computed from time taken by the projectile to travel from one screen to the other. An extension of this concept could be creating a one-dimensional laser screen in the field for the measurement of velocity of fragments produced by a blast. Again, we would require two such screens. Width of screens could be a few tens of metres. Fig. 17 shows a representative set up of laser screens. EFY

*Concluded*